Carbon Nanotubes and their Properties

IN232 Assignment X

Aditi Ajith Pujar

Introduction

As the name suggests, carbon nanotubes (CNTs) are tubular structures of carbon atoms with very small diameter ($\sim$ nm) and large length to diameter ratios. Also visualised as graphite sheets folded along an axis (as the C atoms are in a honeycomb lattice), they are quasi 1D systems. They could be single walled (SWCNTs) or have multiple walls (MWCNTs)

Despite seeming deceptively simple (1D and made of only C atoms), they show an astonishing variety of properties - varying from being highly conducting to having semi-conductor behaviour just by tweaking their geometry. They are the strongest material known in terms of tensile strength and stiffness[1] and they are expected to have wide-ranging applications. We take the first steps toward making sense of any material: understanding its geometry, electronic and phonon band structure and then see how some of its properties come about.

1 Structure of CNTs

1.1 Lattice Geometry

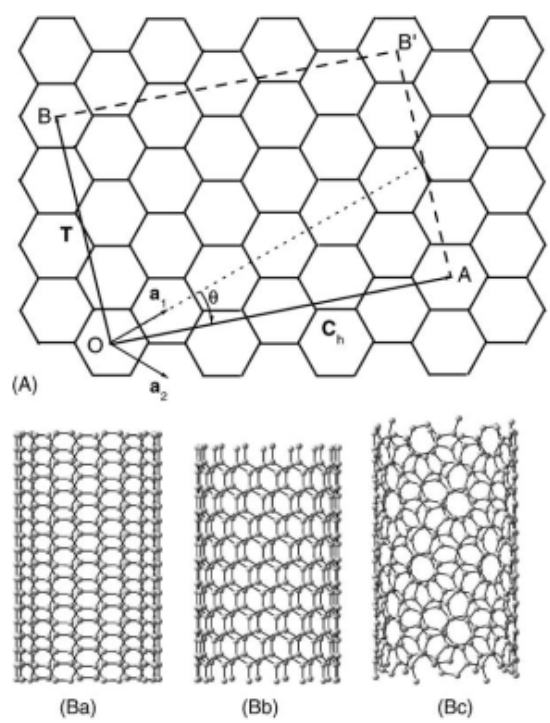

Figure 1: A. Visualisation of CNTs as rolling up of graphite sheets along chirality vector C_h B. Different possible configurations.[3]

As seen in image, a nanotube configuration is defined by some choice of C_h (*chirality vector*, circumferance of tube), T (*translation vector*, length of tube) and θ (*chirality angle*): For $\vec{a_1} = a\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ and $\vec{a_2} = a\left(\frac{\sqrt{3}}{2}, \frac{-1}{2}\right)$ (primitive vectors of honeycomb lattice, a=2.46Å)
If,

$$\boxed{\vec{C_h} = n\vec{a_1} + m\vec{a_2}} \tag{1}$$

then,

$$\boxed{\vec{T} = t_1\vec{a_1} + t_2\vec{a_2}} \tag{2}$$

where

$$\begin{aligned} t_1 &= \frac{2m+n}{d_r}, \\ t_2 &= -\frac{2n+m}{d_r}, \\ d_r &= \text{Greatest Common Divisor of } (2m+n), (2n+m) \text{ and} \end{aligned}$$

$$\vec{C_h} \cdot \vec{T} = 0$$

The number of primitive unit cells per CNT unit cell, N is therefore given by:

$$\boxed{N = \frac{|\vec{C_h} \times \vec{T}|}{|\vec{a_1} \times \vec{a_2}|} = 2\frac{\sqrt{(n^2+m^2+nm)(3m^2+3n^2+5nm)}}{d_R}} \tag{3}$$

Now, nanotubes can be classified on the basis of their (n, m).
A nanotube with chirality of $(n, 0)$ is called a *zig-zag* nanotube. One with chirality of (n, n) is called an *armchair* nanotube and a nanotube without the two mentioned chiralities, is called a *chiral* nanotube.

Reciprocal Lattice:

In the reciprocal lattice of graphite, with basis $\vec{b_1} = \left(\frac{2\pi}{\sqrt{3}a}, \frac{2\pi}{a}\right)$ and $\vec{b_2} = \left(\frac{2\pi}{\sqrt{3}a}, \frac{-2\pi}{a}\right)$:

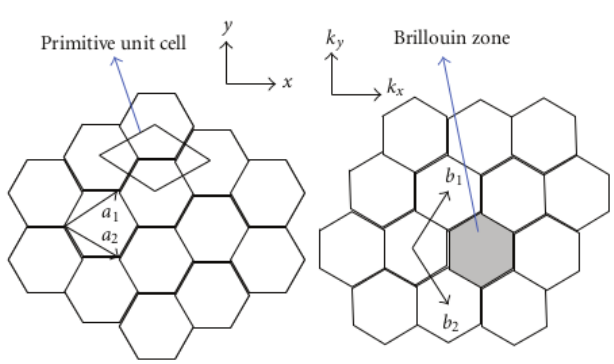

Figure 2: LHS: Honeycomb Lattice of Graphite. RHS: Recirpocal Lattice of Graphite, first Brillouin Zone is shaded [1]

$\vec{k}_\perp$ and $\vec{k}_\parallel$ are the reciprocal vectors of C_h and T, i.e.

$$\begin{aligned} C_h \cdot \vec{k}_\perp &= 2\pi \qquad & C_h \cdot \vec{k}_\parallel &= 0 \\ \text{and} \qquad C_h \cdot \vec{k}_\perp &= 0 \qquad & C_h \cdot \vec{k}_\parallel &= 2\pi \end{aligned}$$

This gives us

$$\vec{k_\perp} = \frac{1}{N}(-t_2\vec{b_1} + t_1\vec{b_2}) \tag{4}$$

$$\vec{k_\parallel} = \frac{1}{N}(m\vec{b_1} - n\vec{b_2}) \tag{5}$$

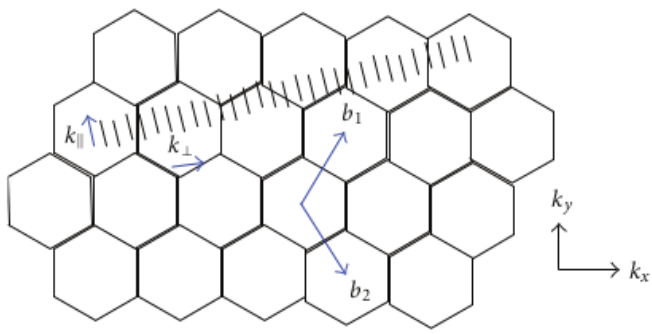

Figure 3: Continuously varying $k_\parallel$ and discretised $k_\perp$ marked in the reciprocal lattice[1]

Due to the periodic (Born-Von Karman) boundary conditions enforced on both, C_h and T, their k-values are quantised as

$$\Delta v = \frac{2\pi}{|v|} \text{ where } v = C_h, T$$

$\implies$ $k_\perp$ is quantised to have N k-points in the first Brillouin Zone.
However, as we expect length to $\to \infty$
$\implies$ $k_\parallel$ is as good as continuous and is the "only reciprocal vector" as we would expect a 1D system to only have one.
We can therefore relate the dispersion relations of the CNTs to the well studied graphite system, by making the "zone folding approximation": Defining

$$\vec{X_k} = \left(s\vec{k}_\perp, \frac{\vec{k_\parallel}}{|\vec{k_\parallel}|}k \right) \text{, where } s = 0, 1, \ldots N-1 \tag{5}$$

We have

$$E_{CNT}(\vec{k}) = E_{graphite}\left(\vec{X_k}\right) \tag{6}$$

$$\text{and } \omega_{CNT}(\vec{k}) = \omega_{graphite}\left(\vec{X_k}\right) \tag{7}$$

This can be interpreted as taking the projection of the graphite dispersion relations in its 2D BZ onto k-vectors along the CNTs' 1D BZ as seen below.

1.2 Electronic Dispersion:

Tight Binding Model in Diatomic Basis of Graphite:

As graphite with its honeycomb lattice has 2 C atoms per primitive unit cell, each unit cell has 2 orbitals, ϕ_1 and ϕ_2. These could be the p_z π orbitals of the $sp-2$ hybridised carbons.

$$\text{Wave function of a unit cell, } \phi = c_1\phi_1 + c_2\phi_2 \tag{8}$$

$$\text{Bloch's Theorem} \implies \psi(x) = \sum_R e^{i\vec{(k)}\cdot\vec{R}}\phi(x-R) \tag{9}$$

Now,

$$\langle\phi_1(x)|\, H\, |\psi(x)\rangle = E\,\langle\phi_1(x)|\,|\psi(x)\rangle = Ec_1$$
$$\langle\phi_2(x)|\, H\, |\psi(x)\rangle = E\,\langle\phi_2(x)|\,|\psi(x)\rangle = Ec_2$$

And by tight binding approximation, defining

$$\langle\psi(x)|\psi(x)\rangle = \epsilon_0$$
$$\text{Hopping term, } \langle\psi(x)|\psi(x+\vec{R})\rangle = t$$
$$\vec{R} = R_1, R_2, R_3 \text{ as each point has 3 nearest neighbours.}$$

Remaining terms in the sum (9) are 0. Solving this gives us

$$E = \epsilon_0 \pm t'\sqrt{3 + \cos\vec{k}\cdot\vec{a_1} + 2\cos\vec{k}\cdot(\vec{a_1}-\vec{a_2})} \tag{9}$$

Or

$$\boxed{E_{2D} = \epsilon_0 \pm t'\sqrt{3 + \cos\frac{a}{2}(\sqrt{3}k_x + k_y) + 2\cos\frac{a}{2}(\sqrt{3}k_x - k_y) + 2\cos ak_y}} \tag{9}$$

Visualising the above with $k_x,\ k_y \epsilon \left(\frac{-\pi}{a}, \frac{\pi}{a}\right)$:

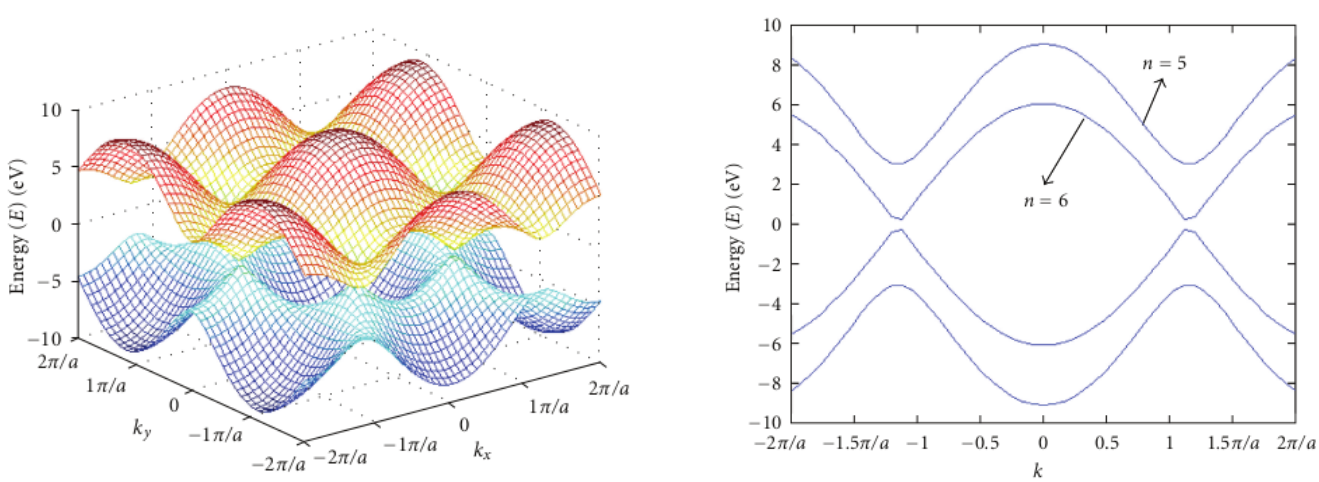

Figure 4: LHS. The energy is localised in the same hourglass shapes of $\pi-$orbitals, around a hexagonal ring and at the centre.[1] RHS. Band structure for metallic $(6,0)$ and semiconductive $(5,0)$ CNTs seen. (The way to obtain them is described in the next subsection).[1]

Now, knowing the dispersion in the first BZ and recalling (5) and (6): For each k-point of $k_\perp$, we project the dispersion E_{2D} along $k_\parallel$ to get the 1D dispersion relations. $\implies$ each k point will have 2N energies: As s can take up to N values in (5), and for each value, X_k, there are two corresponding energies $E_\pm$ as seen from (9). This is also to be expected as there are N lattice points in 2 atom basis, in the CNT unit cells.

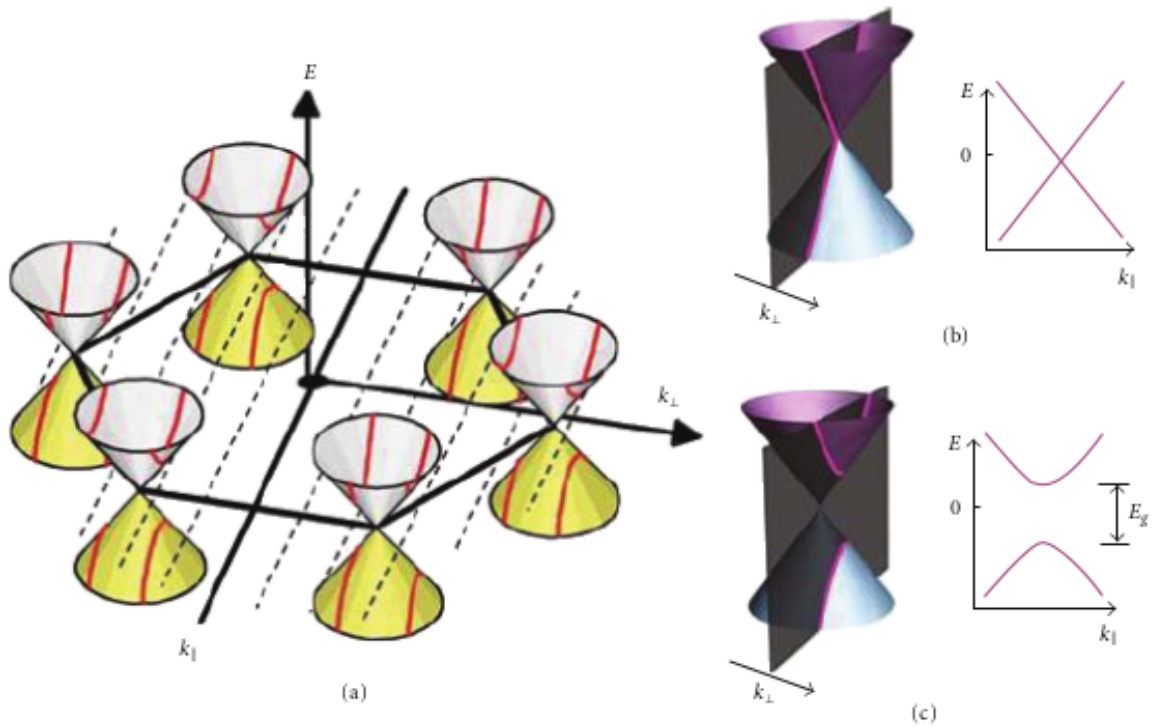

Figure 5: (a) In the 2D 1st Brillouin Zone of Graphite, the dotted lines parallel to $k_\parallel$ axis are the different discrete values taken on by $k_\perp$. The projection is taken onto planes of continuous variable $k_\parallel$ for (b) $(n, m) = (6, 0)$. With ~ 0 band gap, it appears metallic (c) $(n, m) = (5, 0)$. With finite but small band gap, it is semiconductive[1]

Thus, along with applying the same analysis for different families of nanotubes, (Pakhesal et al.)[2] also varied the two physical parameters that define a CNT: r, radius and θ, the chiral angle, for a thorough understanding:

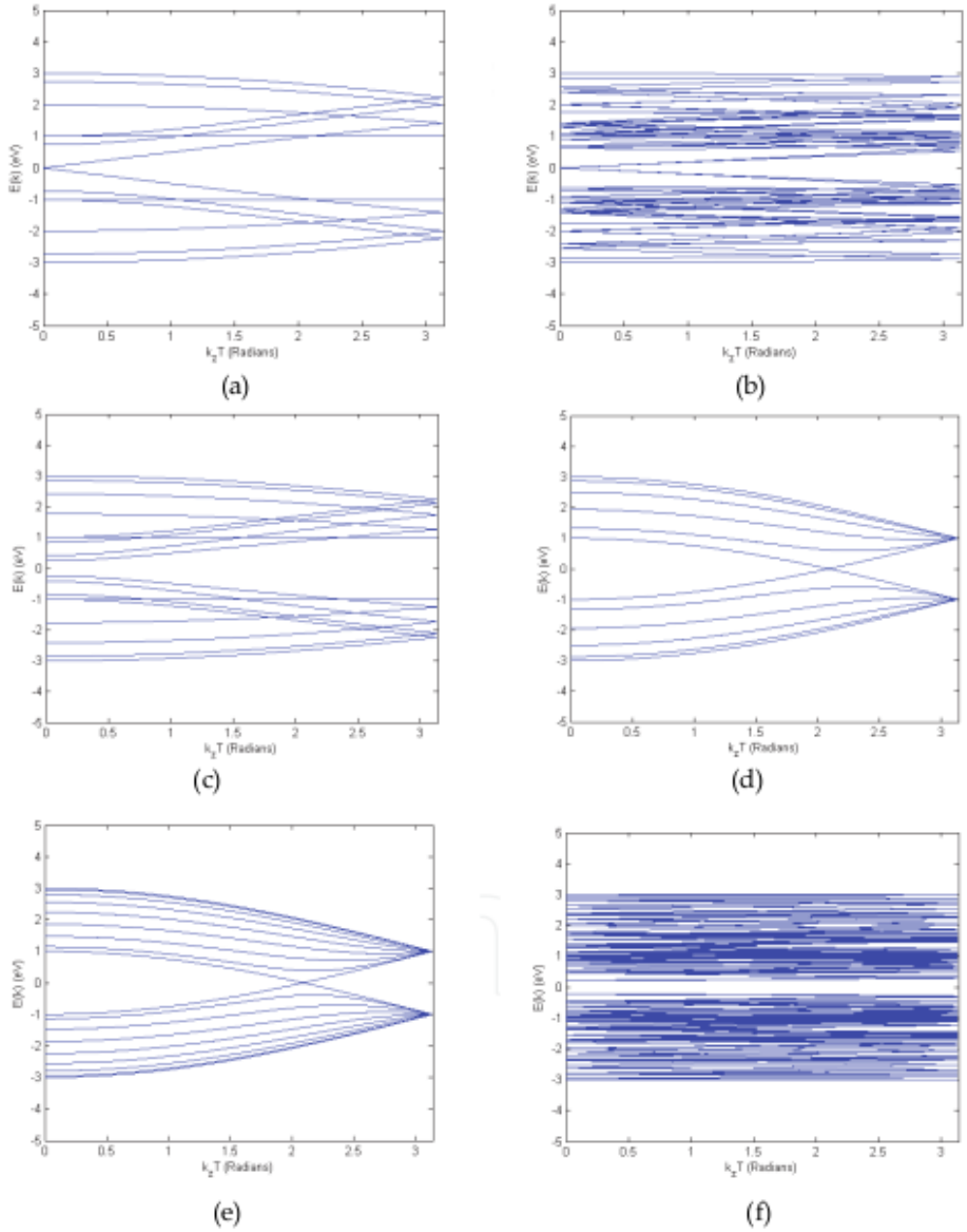

Figure 6: Full band structures ($E\ v/s\ kplots$) for (a) $(6,0)$ (b) $(6,3)$ (c) $(8,0)$ (d) $(5,5)$ (e) $(8,8)$ (f) (5,4).[2]

Characteristics Seen:

1. In Fig 6., unlike in Fig 5(RHS)., all the entire band structures (with as many as $2N$ bands) are plotted.

   (a) (a), (c) with chiral vectors (6,0) and (8,0) respectively are zigzag nanotubes. Similarly, (e), (d) with vectors (5,5) and (8,8) are armchair nanotubes. We can distinctly identify families, as their band structured are qualitatively similar.

   (b) By analysing (9), (5) we can conclude that for a chiral vector (n, m), if $(n-m) \mod 3 = 0$ $\implies$ band gap $E_g \approx 0$
   Thus, all armchair nanotubes with $n = m$ satisfy this condition and therefore have very high conductance. They are metallic to semimetallic.

(c) From (3) We see that, $N\ \alpha\ n,\ m$ to some extent, and greater the N, richer the band structure. This is not a direct, explicit dependance due to the factor of g_R in the denominator, but in Fig 6., the generic trend of increasing number of bands with increasing chiral vectors is evident.

2. The effects of r and θ: it is seen that

   (a) Radius has considerable effect on band gap ΔG and

$$\Delta r\ =\ 0.888nm\ \Longrightarrow\ \Delta G\ =\ 0.025eV$$
$$\Delta G\ \sim \frac{1}{r}$$

   (b) Change of θ has very little effect on band gap. Only upon a deviation $\sim 21.78\deg C$, is $\Delta G\ \sim\ 0.032eV$ seen. Even slightly smaller $\Delta\theta$ caused deviations in band gap an order of magnitude lesser (10^{-4})

1.3 Phonon dispersion:

As seen in (7),

$$\boxed{\omega_{CNT}(\vec{k}) = \omega_{graphite}\left(s\vec{k}_{\perp}, \frac{\vec{k}_{\parallel}}{|\vec{k}_{\parallel}|}k\right)\ \text{, where } s = 0, 1, \ldots N-1} \tag{9}$$

We can see yet again that there will be 2N energies for every k-point in the dispersion relation, it is built into the form of the folding approximation.

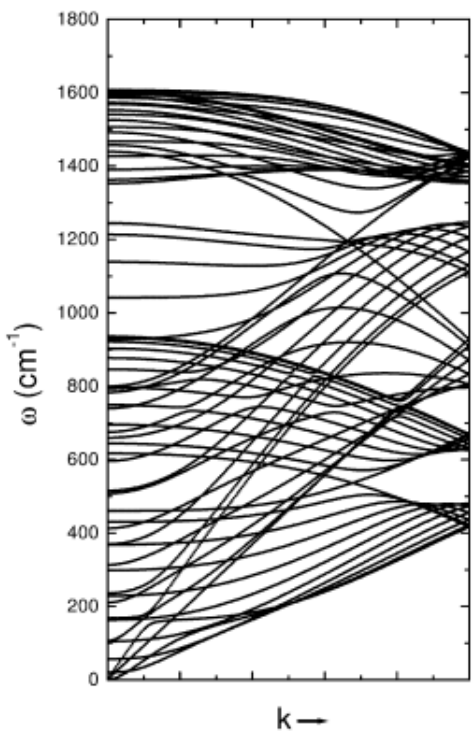

Figure 7: Phonon dispersion calculated for (10,10) CNT. We can see quadratic disperson of transverse branches clearly near centre of BZ. [3]

4 acoustic branches are seen: one longitudinal, two transverse and one twist.

However, there are some discrepancies which show why the zone folding approach is only an approximation, albeit a fair one.

SHORTCOMINGS:

1. At $k = 0$ acoustic phonons:

   (a) Two translational phonons with in-plane atomic displacements/atomic displacement perpendicular to the tube axis are not eigenmodes of the graphite sheet. To obtain these two modes by zone-folding, we need to obtain it by perturbative methods.

   (b) A translational phonon will have out-of-plane atomic displacements: upon deformation into 1D cylinder, the phonon will correspond to a "breathing mode", in which all atoms vibrate uniformly along the radial direction, the so-called radial- breathing mode (RBM).

2. Does not take the strain on the lattice due to folding into account $\implies$ -the hybridization between the sp^2 and p_z orbitals is small.

2 Properties

2.1 Electrical Properties

The most important property to understand would be the conductivity, its dynamics and temperature dependance and so on. However, as the system is 1D:

The electrons cannot avoid each other, as their motion is highly constrained. Moreoever, electron-electron interactions themselves are much stronger because screening effect is quite poor in 1D. Thus, as their motion becomes highly correlated and collective, gross level nearly free electron models like Sommerfeld's will not help. The motion of the electrons can be described as flow of a Luttinger fluid. [4]

2.2 Thermal Properties

Phonons contribute much more predominantly to specific heat and conductivity than electrons.

2.2.1 Specific heat

1. C_v between 20-300K has linear dependance on temperature this is because, in 1D, the phonon specific heat has linear dependance in T.

$$
\begin{aligned}
\sum_n &\sim \int d\frac{\omega}{v} \\
g(\omega) &\sim \int d\omega && (9)\\
E &\sim \int d\omega g(\omega) n_B(\beta\hbar\omega) && (10)\\
E &\sim T^2 && (11)\\
&\Longrightarrow \boxed{C_v = \frac{\partial E}{\partial T} \sim T} && (12)
\end{aligned}
$$

2. Bundles of SWNTs showed deviations at low temperatures from SWNTs which could be explained by a 2 band Debye model due to weak interactions between the nanotubes.

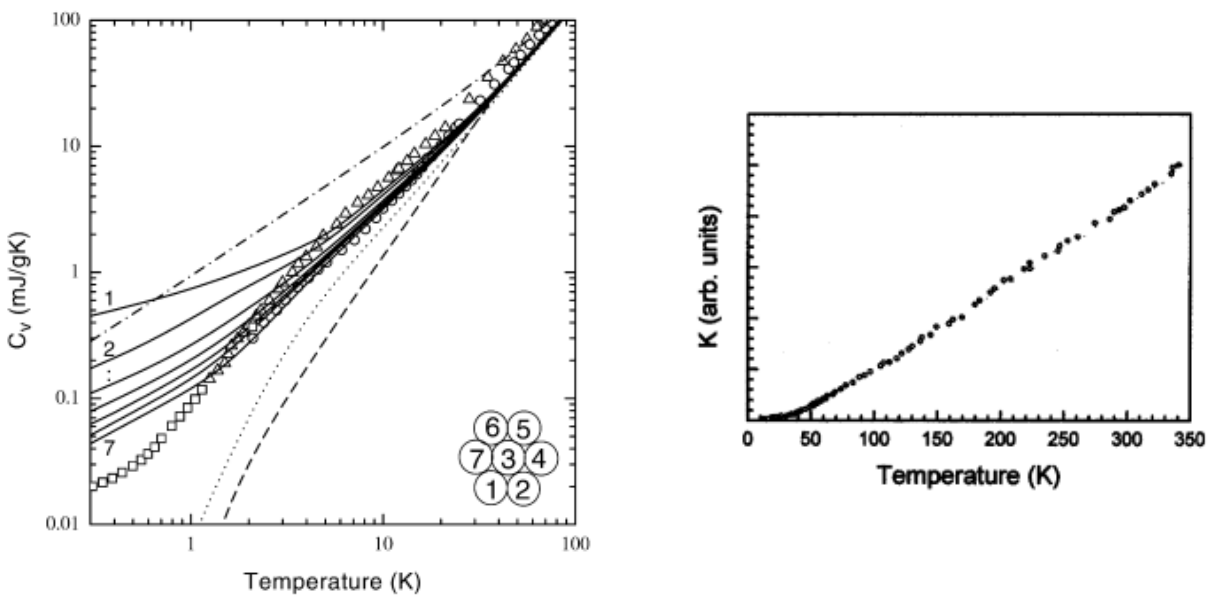

Figure 8: LHS. Specific heat temperature dependance showing a predominantly linear dependence, though some 2D-3D like behaviour comes through at higher T. RHS. Thermal conductivity K dependance on T

2.2.2 Thermal conductivity:

In general, between 4-300K,

$$\kappa \sim T^2$$

This is similar to graphene. It is noteworthy that there is no maxima observed here $\Longrightarrow$ no phonon Umklapp scattering processes.

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Thus, with many unexplored avenues of potential applications, there is much reason to believe carbon nanotubes along with other "carbon based materials are likely to be a major field in the twenty first century technology".[3]

3 References

[1]Davood Fathi, "A Review of Electronic Band Structure of Graphene and Carbon Nanotubes Using Tight Binding", *Journal of Nanotechnology*, vol. 2011, Article ID 471241, 6 pages, 2011. https://doi.org/10.1155/2011/471241

[2]Rahim Ghayour and Pakkhesal (July 27th 2011). Electronic Band Structure of Carbon Nanotubes in Equilibrium and None-Equilibrium Regimes, Electronic Properties of Carbon Nanotubes, Jose Mauricio Marulanda, IntechOpen, DOI: 10.5772/17556. Available from:https://www.intechopen.com/books/el properties-of-carbon-nanotubes/electronic-band-structure-of-carbon-nanotubes-in-equilibrium-and-none-equilibrium-regimes

[3]Valentin N Popov, Carbon nanotubes: properties and application, *Materials Science and Engineering*: R: Reports, Volume 43, Issue 3, 2004, Pages 61-102, ISSN 0927-796X, https://doi.org/10.1016/j.mser.2 (http://www.sciencedirect.com/science/article/pii/S0927796X03001268)

[4]https://en.wikipedia.org/wiki/Carbon_nanotubeMechanical

[5]https://en.wikipedia.org/wiki/Luttinger_liquid

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